Active Learning-Guided Protein Engineering: From Receptor Orthogonality to Biosensor Design

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This proposal covers two projects I have been working on this semester, both built around the same idea: using machine learning to navigate sequence-function landscapes. In the first project, I computationally engineered a plant immune receptor to selectively recognize synthetic peptides while rejecting native ones, using active learning and AlphaFold3 scores as a proxy for binding fitness. In the second, I use machine learning to predict fluorescence biosensor dynamic range from their sequence, where fitness comes from experimental sort-seq data.

<u>Directed Evolution of the SCOOP-MIK2 System</u>

Background and Research Question

Plant immune systems have evolved sophisticated pattern recognition receptors to detect molecular signals of danger or fungal invasion. One of these is MIK2, a leucine-rich repeat receptor-like kinase. MIK2 is essential for identifying SCOOP peptides, a family of endogenous damage-associated molecular patterns. These 13-15 amino acid long peptides contain a conserved SxS motif and activate immune responses in *Arabidopsis thaliana* and related species upon recognition by MIK2[1]. However, the evolutionary success of plant immunity has been tempered by an equally adaptive adversary: the pathogens themselves. Several fungal pathogens, notably *Fusarium oxysporum*, have evolved SCOOP-like peptides that interfere with host signaling. In response, plants have evolved the MIK2 receptor to detect a few of these peptides and trigger defense responses[2,3,4]. This phenomenon illustrates a molecular arms race, where plants continually refine receptor sensitivity, while pathogens evolve convergent or deceptive molecular patterns that can evade or alter immune recognition.

Computational directed evolution offers us a much faster way to evolve peptides and receptors as compared to nature's slower evolutionary pace. The main question of this project, therefore, is: Can we computationally evolve the SCOOP-MIK2 system to create an orthogonal receptor-ligand pair? A receptor with the desired properties would serve as a programmable plant immune circuit that responds only to the synthetic trigger peptides from my previous work, while rejecting native SCOOPs. Addressing this question not only sheds light on the structural and evolutionary basis of receptor-ligand specificity but also establishes a generalizable framework for creating synthetic immune primers for crops.

Methodology and Interim Results

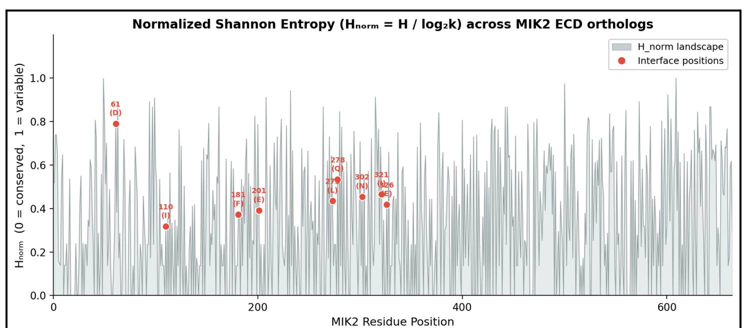

Fig. 1 | Normalized Shannon entropy (H_norm = H / $\log_2 k$) across 52 MIK2 ECD orthologs. Red circles mark the nine interface residues selected for mutagenesis, labeled by residue number and wild-type amino acid.

Receptor Selection

The first step in creating an orthogonal receptor is choosing which MIK2 ortholog to use as the base starting receptor. For this, I modelled all the MIK2 clade orthologs and paralogs against the top two peptides ALDE1 and ALDE2, created using the active learning directed evolution framework[5] from my previous work, and two native SCOOPs (SCOOP10B and SCOOP12) using AlphaFold3[6]. A delta fitness score was used to choose the receptor with the most natural ALDE selectivity using the ipTM metric: $\Delta fitness = max(ipTM^{ALDE}) - max(ipTM^{SCOOP})$. *C.rubella* MIK2 *(Carub0001s3337*) had the highest delta fitness and thus became the engineering skeleton. Starting from the ortholog with the strongest existing selectivity bias makes every subsequent mutation more likely to be selective to ALDEs over SCOOPs. To select the mutation sites, I used two sequential filters to define the mutagenesis space. First, AlphaFold3 structures of CrMIK2-ALDE1 were analyzed for heavy atom contacts within 5 Å of the peptide (this cutoff captures all non-covalent interactions: H-bonds, salt bridges, van der Waals contacts, and π-π stacking), returning 43 interface receptor residues. Second, MSA of all MIK2 orthologs (Fig. 1) identified which of those 43 positions tolerate natural substitution. This intersection gave us nine candidate sites [61, 110, 181, 201, 273, 278, 302, 321, 326] that are both structurally relevant and evolutionarily permissive.

Epistasis Analysis

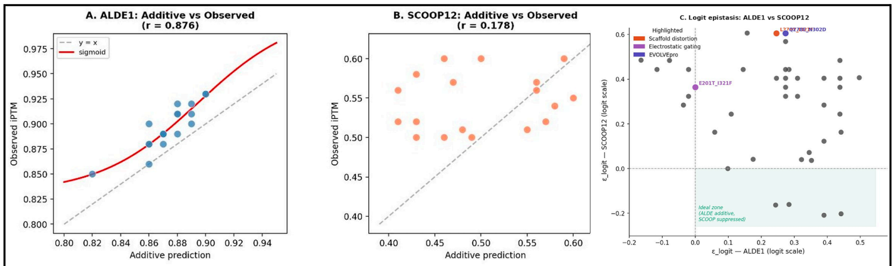

Fig. 2 | Epistasis analysis across CrMIK2 double mutants. **(A)** Additive model prediction vs. observed ALDE1 iPTM across double mutants. **(B)** Additive model prediction vs. observed SCOOP12 iPTM **(C)** Logit-transformed epistasis for ALDE1 vs. SCOOP12; the shaded ideal zone (bottom right) marks mutants with positive ALDE epistasis and negative SCOOP epistasis. Highlighted mutants represent the three top candidates identified downstream.

Following this, I modelled all possible single amino acid mutants at these interface residues with ALDE1 and SCOOP12. I used those ipTM scores to build an additive null model to test the hypothesis that individual interface residues contribute independently to binding selectivity. For ALDE1 binding, the hypothesis held up reasonably well (r = 0.876, Fig. 2A). Every tested double mutant exceeded the additive prediction, indicating that combining two mutations is better for ALDE1 binding than the sum of their parts. However, for SCOOP12 suppression, this hypothesis failed (r = 0.178). The scatter is not noise; it shows non-additive interactions between sites (Fig. 2B). This epistatic interaction in SCOOP12 binding means that screening single mutants is insufficient to predict which double mutants will suppress SCOOPs. In order to quantify this epistatic interaction, I used a logit-transformed epistasis metric[7] for ALDE1 vs SCOOP12:

$$\varepsilon_{logit} = f'ab - f'a - f'b + f'WT$$

The fitness score f was logit transformed, as naive epistasis metrics on bounded scales produce artifacts. Fig. 2C shows the predicted "ideal zone", where mutants are expected to show positive ALDE epistasis and negative SCOOP epistasis. The three highlighted mutants, which showed the strongest downstream performance, do not fall in this zone. I also evaluated the four peptides in this ideal zone to see if they were selective; however, AlphaFold 3 modeling revealed that two do not suppress SCOOP10B (D61F_I321D, Q278F_I321M) and another fails to bind with ALDE2 (N302F_I321D). This contradiction suggests that the best selectivity does not always come from the most epistatically synergistic combination, but rather from mutations whose mechanisms suppress SCOOP recognition. This experiment revealed a limitation of the additive null model approach, which is what motivated the use of a more complex model. Nevertheless, the top predictions from this model were further tested alongside other candidates.

EVOLVEpro for double mutant generation

In order to capture non-linear interactions, I adopted the EVOLVEpro[8] framework. A random forest regression model was trained on ESM-2[9] protein language model embeddings for 171 single mutant sequences across the 9 interface residues. This model was then used to rank all 12,996 possible double mutants using their respective embeddings. Top predictions from the EVOLVEpro pipeline, via greedy selection, were selected as candidates for AlphaFold3 validation. Of the 20 EVOLVEpro nominated double mutants validated by AlphaFold3, 18 met the selectivity threshold (ALDE1 ipTM ≥ 0.80, SCOOP12 ipTM < 0.80); however, this is not a generalized result, and hence these mutants required further testing across all the peptides. After testing all the double mutants drawn from both the EVOLVEpro nominations and the additive model's top predictions against all SCOOP and ALDE peptides, three stood out as the top candidates, each characterised by a different mechanism:

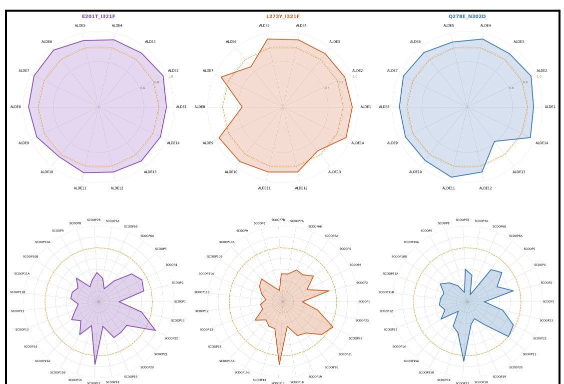

Fig. 3 | ALDE binding fingerprints and SCOOP rejection profiles for the three top double mutant candidates. The dashed inner ring marks the 0.80 ipTM threshold. Top row shows binding across all 14 ALDE peptides per mutant, bottom row shows binding across all 27 SCOOP peptides (polygons remaining inside the dashed ring indicate SCOOP rejection).

E201T_I321F, Charge Removal: AlphaFold3 model of CrMIK2-SCOOP12 predicts a bidentate salt bridge between E201(Glu) and the Arg residue of SCOOP12. Replacing the negatively charged glutamic acid with threonine removes a charge contact critical for SCOOP binding. This mutation produces the strongest SCOOP suppression in the campaign (SCOOP12 ipTM = 0.49) while maintaining a mean ALDE ipTM of 0.91.
L273Y_I321F, Scaffold Distortion: L273 mutations destabilize SCOOP binding regardless of the second mutation. L273 shapes the LRR backbone that forms the SxS-binding canyon. This is consistent with the hypothesis that L273 substitutions distort canyon geometry in a way that penalizes the rigid SxS motif of native SCOOPs more than it penalizes ALDEs, which have more positional flexibility outside the motif.
Q278E_N302D, EVOLVEpro nomination: Neither position (Q278 or N302) stands out in the single mutant data, and the sequence-fitness correlation is unclear. The additive model doesn't predict that this combination would work, but the EVOLVEpro model identified it using a feature space that the additive model can't access.

These three mutants show broad ALDE recognition, maintaining ipTM ≥ 0.80 across all the 14 designed ALDE peptides from my previous work (designed with the active learning directed evolution framework), while suppressing native SCOOP binding to below the selectivity threshold across the 27 natural SCOOP peptides (Fig. 3).

Classification of Dynamic Range of Biosensors

My second project shares the same underlying challenge as the SCOOP-MIK2 evolution: the sequence space is too large to screen exhaustively. In the MIK2 project, I navigated that problem using structural predictions of the mutants as a fitness proxy. Here, I collaborate with a postdoctoral researcher in the lab and contribute computational modeling to her work on fluorescence biosensor design. These biosensors are derived from the transporter proteins SWEETs, where short linker peptides are present between fluorescent proteins (cpFP) and the transporter to couple conformational change in the transporter to fluorescence output. Because these biosensors are chimeric proteins and linked by only a few amino acids, their behavior cannot be predicted solely from their sequence. Therefore, our lab used a high-throughput sort-seq workflow[12], where 103,718 linker variants were expressed in yeast, sorted into four logarithmic fluorescence bins using FACS under glucose and starved conditions, and used NGS to obtain read counts per bin. From these read counts, dynamic range (DR) was estimated per variant, and I previously implemented a maximum likelihood estimation in the $\log_{10}$ space to model the fluorescence intensity. Following this, I started by correcting the unequal sequencing depth across bins with a per-bin scaling factor, d_j:

$$dj = \sum bj / \sum rj$$

where b_j is the number of cells physically sorted into bin j, and r_j is the number of sequencing reads from bin j. This rescales read counts to reflect actual cell proportions rather than sequencing artifacts from over or under sequenced bins. The final dataset contained 1,424 variants with DR ranging from −0.74 to 7.63.

To predict the dynamic range of these biosensors, I framed the problem as a binary classification task. Variants with DR ≥ 1 (a 2-fold change) were labeled as High. As MLE estimates are noisier for low-frequency variants, I swept read-count thresholds from 20 to 200 reads to assess how measurement reliability trades off against sample size. In the full dataset (n = 1,424), 38% of variants were High. By cutoff ≥ 100, that dropped to 16% across only 188 variants. Evaluating model performance at different cutoffs allowed us to identify where classifier performance was driven by the signal in the data. Each variant was encoded using VHSE physicochemical descriptors (8 features per residue, 48 dimensions total for the 6 amino acid linker). ESM-2 embeddings of the full 501-residue chimera sequence were generated, and the left and right linker embeddings were concatenated and reduced using PCA (50 PCs capturing ~90% of variance), giving us a combined 98-dimensional feature vector for the linkers. Three classifiers were evaluated under 5-fold cross-validation (Fig. 4A) with class_weight='balanced' to correct for imbalance: Random Forest, Logistic Regression, and SVM with RBF kernel. F1 score, AUC-ROC, and precision-recall curves were used as primary metrics for evaluation; accuracy was not, since it inflates as the class imbalance worsens.

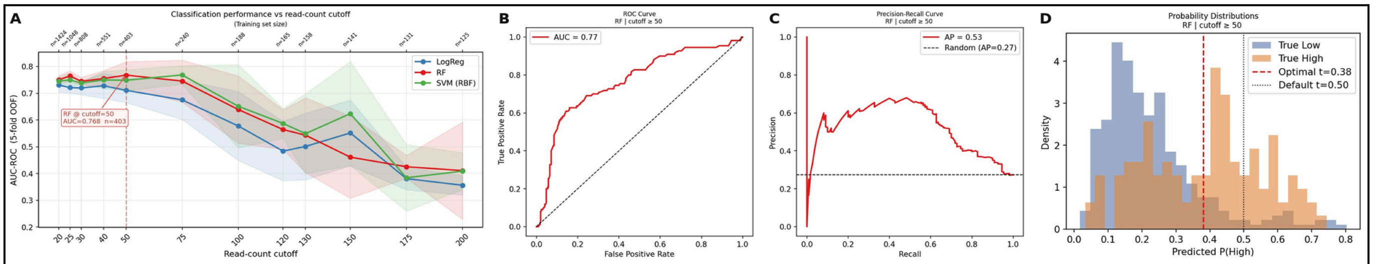

Fig. 4 | Random Forest classifier performance for biosensor dynamic range prediction. **(A)** AUC-ROC across read-count cutoffs for three classifiers; the dashed line marks the selected cutoff of 50 (n = 403, AUC = 0.768). **(B)** ROC curve for the RF model at cutoff ≥ 50. **(C)** Precision-recall curve compared to random baseline. **(D)** Probability distributions for true high and true low dynamic range variants; optimal decision threshold t = 0.38 recovers additional true positives relative to the default t = 0.50.

The Random Forest model at cutoff 50 showed the best performance. With an AUC of 0.768 (Fig. 4B) and an F1 score of 0.74, this model shows reasonable classification power. Precision for the low DR class (0.84) is higher than that of the high DR class(0.63), with an average precision of 0.53 (Fig. 4C). Decision threshold tuning lowered the operating point from the default 0.50 to 0.38, recovering additional true positives at the cost of some false positives. This trade-off is visible in the probability distribution plot (Fig. 4D). Beyond cutoff ≥ 100, performance degraded across all three models as the sample sizes dropped below 200. This work allows us to better understand the correlation between changes in linker peptide sequence and the corresponding fluorescence response of the biosensor.

The Plan Ahead

The current phase of this research demonstrates the feasibility of *in silico* directed evolution for creating orthogonal receptor-ligand pairs. Moving forward, with the inputs and suggestions from my mentor and principal investigator, Dr. Lily Cheung, I foresee three complementary avenues that will define the next phase of this project. First, the iterative part of active learning involves **experimentally testing** the designed peptides and receptors and using that data for further rounds of model training. To gather this data, a ChBE undergraduate student in our lab is currently running experiments on the designed ALDE peptides to test binding using root growth, root browning, and ROS burst assays[4]. We also plan to test the mutated receptor variants with the designed ALDE peptides and native SCOOPs to test for orthogonality. If the assays confirm ALDE peptide binding, that data enters the active learning retraining loop and shifts the fitness landscape for the next round of receptor mutations. If it disconfirms binding, we identify which ipTM threshold is insufficient as a proxy for in vivo activity and recalibrate the scoring function. Either outcome advances the model. Second, **further rounds of predictions** using the EVOLVEpro framework while including AlphaFold 3 validated ipTM scores. Each subsequent round would retrain the random forest on the expanded label set and re-rank the remaining unlabeled doubles to nominate the next set of mutants, and this can also be expanded to triple or higher order mutants. Thereafter, we intend to implement other state-of-the-art models like RFdiffusion3[10] or BindCraft[11] to generate de novo peptides for the designed receptors. Third, I am currently exploring **orthogonality from the peptide** side. I am re-using the active learning framework from my previous work to mutate SCOOP peptides to try to break the evolutionarily conserved SxS motif, by exploring substitutions at the S7 position. This opens up a potential second axis of orthogonality.

For the biosensors project, the results show that sequence-based classification of biosensor dynamic range is feasible but constrained by the diversity and reliability of the training data. Moving forward, I intend to use the EVOLVEpro framework I applied for ranking receptor mutants to rank biosensor linker sequences across the 20^6 sequence space. Top candidates will be experimentally tested, and these results can be fed back as new labeled data to retrain the model, closing the active learning loop and improving prediction quality with each round.

Coming from a Computational Biology background, I have always been passionate about leveraging AI to solve problems with broader, practical applications. The protein engineering methods I have implemented, like active learning, structure based fitness scoring, and epistasis analysis are directly transferable to human drug targets, and that is exactly where I intend to take them. I plan to apply these computational skills during my summer internship in the Translational Safety & Risk Sciences group at Amgen. I'm confident that this research will enable me to not only analyze biological systems better but also build AI-driven drug discovery pipelines at the forefront of therapeutic development.

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